# Model Revisions in General Equilibrium

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# motivation

Subjective expectations of returns differ from econometric predictions and forecasts

- systematic forecast errors
- subjective-objective gap varies over time

1. What are the equilibrium consequences of subjective models?

- what if investors revise their models (are subjective models fixed)?
- what if investors have conflicting models (disagree with econometrician or each other?)

2. What can we learn about the economy from dynamics of expectations + errors?

# this paper

A tractable model with dynamic and heterogeneous models

- uncertainty across models + model revisions $\longrightarrow$ dynamic risk limits
- aggregation in general equilibrium
- clean identification of subjective beliefs and objective data-generating process
- clean identification of model revision shocks and redistribution

# outline

Portfolio choice with dynamic model misspecification concerns and revisions

- relation to value-at-risk constraints

Aggregation:

- asset prices, risk premia depend on wealth distribution through single scalar
- financial shocks (model revisions) do not induce redistribution, real shocks do
- wealth distribution driven by "representative" agent's preferred model

Application: dynamics of equity premium in the US, 1987-2021

- identification of revisions (70%) vs redistribution from subjective and objective expectations
- why subjective premia are less volatile and cyclical than objective
- how 2000 was different from 2008 (what if a new 2000 soon?)

# literature

Robustness concerns:

- Gilboa and Schmeidler (1989), Hansen and Sargent (2001)

Related preferences and portfolio constraints:

- Danielsson, Shin, and Zigrand (2012), Adrian and Boyarchenko (2018), Hofmann, Shim, and Shin (2022), Coimbra (2020), Coimbra and Rey (2024), Gromb and Vayanos (2002), Gromb and Vayanos (2018), Vayanos and Vila (2021), Gourinchas, Ray, and Vayanos (2022), Greenwood, Hanson, Stein, and Sunderam (2023), Itskhoki and Mukhin (2021), Kekre and Lenel (2024), Kekre and Lenel (2025), Adrian and Shin (2010), Adrian and Shin (2014), Coimbra, Kim, and Rey (2022), Barbiero, Bräuning, Joaquim, and Stein (2024)...

Objective and subjective beliefs:

- Piazzesi, Salomao, and Schneider (2015), **Bhandari, Borovička, and Ho (2025)**, **Szőke (2022)**, **Nagel and Xu (2023)**

# A model of model revisions

# environment

State $x_t$ driven by $\{Z_t\}_{t\geq 0}$, a Brownian motion under the physical measure $\mathbb{P}$:

$$dx_t = \mu_x(x_t)dt + \sigma_x(x_t)dZ_t$$

Risk-free instant-maturity bond pays $r(x_t)$, risky assets with excess returns $dR_t$:

$$dR_t = \mu_R(x_t)dt + \sigma_R(x_t)dZ_t$$

Agent's model: two different processes $\{B_t, Z_t\}_{t\geq 0}$

$$dx_t = \mu_x(x_t)dt + \sigma_x(x_t)dZ_t$$

$$dR_t = \mu_R(x_t)dt + \sigma_R(x_t)dB_t$$

Moreover, not sure about $\{B_t\}_{t\geq 0}$: alternative measure $\mathbb{Q}$ under which $\{\tilde{B}_t\}$ is Brownian

$$\tilde{B}_t = B_t + \int_0^t h_s ds$$

# minmax problem

Log ratio $m_t$ of likelihood of return data under $\mathbb{Q}$ and $\mathbb{P}$ evolves as

$$dm_t = \frac{|h_t|^2}{2}dt - h_t' d\tilde{B}_t$$

Agent solves a min-max problem with a discipline parameter $\psi_t$

$$\max_{\{c_t,\theta_t\}} \inf_{\mathbb{Q}} \mathbb{E}^{\mathbb{Q}}\left[\int_0^\infty \rho e^{-\rho t}\log(c_t)dt + \int_0^\infty e^{-\rho t}\psi_t dm_t\right]$$

$$dw_t = (r(x_t)w_t - c_t)dt + w_t\theta_t' dR_t$$

- consumption $c_t$, wealth $w_t$
- portfolio shares $\theta_t$ over risky assets (do not sum to 1)

# consumption and portfolio choice

## Proposition 1

Consumption and portfolio choice are

recursive formulation extensions

$$c^*(w_t, x_t) = \rho w_t$$

$$\theta^*(w_t, x_t) = \frac{\psi_t}{\psi_t + 1} \cdot [\sigma_R(x_t)\sigma_R(x_t)']^{-1}\mu_R(x_t)$$

# risk-limit environment

Agent's problem: given a process for $\gamma_t \in [0,1]$, heuristic explanation

$$\max_{\{c_t,\theta_t\}_{t\geq 0}} \mathbb{E}\int_0^\infty \rho e^{-\rho t}\log(c_t)dt$$

$$\text{s.t. } \mathbb{V}_t[\theta_t' dR_t] \leq \gamma_t \mathbb{E}_t[\theta_t' dR_t] \qquad \text{(value-at-risk)}$$

$$dw_t = (r(x_t)w_t - c_t)dt + w_t\theta_t' dR_t$$

**Proposition 2**

A robust agent with a cost parameter $\psi_t$ has the same consumption and portfolio choice as the value-at-risk constrained agent with a multiplier $\gamma_t = \dfrac{\psi_t}{\psi_t + 1}$. technical details

- cannot distinguish beliefs from risk limits (Borovička, Hansen, and Scheinkman (2016))

# relation to literature

Existing alternatives:

- standard robustness setup of Hansen and Sargent (2001) — standard setup

$$\theta^*(w_t, x_t) = \frac{\psi_t}{\psi_t + 1} \cdot [\sigma_R(x_t)\sigma_R(x_t)']^{-1}\mu_R(x_t) + \phi(x_t)$$

- recursive preferences of Kreps and Porteus (1978), Duffie and Epstein (1992) — recursive preferences

$$\theta^*(w_t, x_t) = \gamma_t \cdot [\sigma_R(x_t)\sigma_R(x_t)']^{-1}\mu_R(x_t) + \varphi(x_t)$$

- capping **std**: Danielsson, Shin, and Zigrand (2012), Adrian and Boyarchenko (2018)

$$\theta^*(w_t, x_t) = \lambda(\gamma_t, w_t, x_t) \cdot [\sigma_R(x_t)\sigma_R(x_t)']^{-1}\mu_R(x_t)$$

# Aggregation

# an economy with heterogeneous investors

- agents $i \in \{1, ...n\}$, individual states: parameters $\{\gamma_{it} \equiv \psi_{it}/(\psi_{it}+1)\}$ and wealth $\{w_{it}\}$
- risky assets $j \in \{1, ...k\}$ in fixed supply $\{s_j\}$ priced at $\{p_{jt}\}$, pay dividends $\{y_{jt}\}$
- risk-free instant maturity bonds in zero net supply pay $r_t$
- agents' portfolio shares $\{\theta_{ijt}\}$ translate to holdings $m_{ijt} = \theta_{ijt} w_{it}/p_{jt}$ and $b_{it} = (1 - \theta_{it}' 1_k) w_{it}$

With $\boldsymbol{y}_t = \{y_{jt}\}$, $\boldsymbol{\gamma}_t = \{\gamma_{it}\}$, $\boldsymbol{w}_t = \{w_{it}\}$, aggregate states are $x_t = (\boldsymbol{y}_t, \boldsymbol{\gamma}_t, \overline{\boldsymbol{w}}_t)$, where $\overline{\boldsymbol{w}}_t = \boldsymbol{w}_t$ a.s.

$$d\boldsymbol{y}_t = \mu_y(\boldsymbol{y}_t)dt + \sigma_y(\boldsymbol{y}_t)dZ_t^y$$
$$d\boldsymbol{\gamma}_t = \mu_\gamma(\boldsymbol{\gamma}_t)dt + \sigma_\gamma(\boldsymbol{\gamma}_t)dZ_t^\gamma$$

# equilibrium

Given shocks $\{y_{jt}, \gamma_{it}\}_{t\geq 0}$, an **equilibrium** is a set of adapted processes for prices $\{p_{jt}, r_t\}_{t\geq 0}$ and quantities $\{w_{it}, c_{it}, b_{it}, m_{ijt}\}_{t\geq 0}$ that solve agents' problems with prices taken as given and satisfy

$$\sum_i m_{ijt} = s_j \text{ for all } j$$
$$\sum_i b_{it} = 0$$
$$\sum_i c_{it} = \sum_j s_j y_{jt}$$

# equilibrium characterization

Characterize prices $\boldsymbol{p}(x_t) = \{p_j(x_t)\}$ and $r(x_t)$ as functions of aggregate states: returns

$$d\boldsymbol{p}(x_t) = \mu_p(x_t)dt + \sigma_{p,y}(x_t)dZ_t^y + \sigma_{p,\gamma}(x_t)dZ_t^\gamma$$

Total wealth is exogenous: $\rho \sum_i w_{it} = \sum_j s_j y_{jt}$

Denote $w_t = \sum_i w_{it}$ and define $\{\mu_w(\boldsymbol{y}_t), \sigma_w(\boldsymbol{y}_t)\}$ by $\dfrac{dw_t}{w_t} \equiv \mu_w(\boldsymbol{y}_t)dt + \sigma_w(\boldsymbol{y}_t)dZ_t^y$

Denote wealth shares by $\nu_{it} = \dfrac{w_{it}}{w_t}$ and define the weighted average $\Gamma_t$ and dispersion $\Delta_t$

$$\Gamma_t = \sum_i \nu_{it}\gamma_{it}$$
$$\Delta_t = \sum_i \nu_{it}\gamma_{it}^2 - \left(\sum_i \nu_{it}\gamma_{it}\right)^2$$

# asset prices

## Proposition 3

The interest rate and asset prices solve

corollary

$$r(x_t) = \rho + \mu_w(\boldsymbol{y}_t) - \frac{|\sigma_w(\boldsymbol{y}_t)|^2}{\Gamma_t}$$

$$r(x_t)\boldsymbol{p}(x_t) = \boldsymbol{y}_t + \mu_p(x_t) - \frac{\sigma_{p,y}(x_t)\sigma_w(\boldsymbol{y}_t)'}{\Gamma_t}$$

- only loadings of prices on output shocks $\sigma_{p,y}(x_t)$ matter: loadings $\sigma_{p,\gamma}(x_t)$ do not enter
- $\Gamma_t < 1$ increases the risk premium, decreases interest rate
- regular log utility nested as $\Gamma_t \equiv 1$

# stochastic discount factor

**Proposition 4**

Asset prices satisfy

$$\Lambda_t \boldsymbol{p}(x_t) = \mathbb{E}_t \int_t^\infty \Lambda_s \boldsymbol{y}_s ds$$

Here $\Lambda_0 = 1$ and

$$\frac{d\Lambda_t}{\Lambda_t} = -r(x_t)dt - \frac{1}{\Gamma_t}\sigma_w(\boldsymbol{y}_t)' dZ_t^y$$

# weak aggregation

Wealth distribution enters asset pricing relations through one weighted average

- "representative" agent with $\gamma_t = \Gamma_t$ on all sample paths
- $\Gamma_t$ is not Markov, only complete state $x_t = (\gamma_t, \boldsymbol{y}_t, \boldsymbol{\nu}_t)$ is Markov
- non-trivial wealth distribution dynamics

Next: what can we say about the dynamics of $\Gamma_t$ and the wealth distribution?

# Evolution of states

# representative model

Hypothetical single agent with robustness parameter $\Psi_t = \dfrac{\Gamma_t}{1-\Gamma_t}$

- has access to a single asset: total stock market index with excess return $dR_t$

$$dR_t = \frac{1}{\sum_j s_j p_j(x_t)} \sum_j s_j p_j(x_t) dR_{jt}$$

- chooses an alternative probability measure $\mathbb{Q}_t$ out of robustness concerns
- forecast error process $dW_t \equiv dR_t - \mathbb{E}^{\mathbb{Q}_t}[dR_t]$

# representative forecast error

**Proposition 5**

The forecast error process $dW_t$ in the investor's preferred model is

$$dW_t = \underbrace{\frac{1-\Gamma_t}{\Gamma_t}}_{=\Psi_t^{-1}} |\sigma_w(\boldsymbol{y}_t)|^2 dt + \sigma_w(\boldsymbol{y}_t) dZ_t^y$$

- positive drift if there are robustness concerns on aggregate
- no contribution of financial shocks to forecast errors
- no drift in the nested log utility case

# dynamics of wealth distribution

Define agent $i$'s leverage as the total risky share of her portfolio: $\lambda_{it} \equiv \sum_j \theta_{ijt}$

**Proposition 6**

Agent $i$'s leverage is $\lambda_{it} = \dfrac{\gamma_{it}}{\Gamma_t}$. Her wealth share $\nu_{it}$ evolves as

$$\frac{d\nu_{it}}{\nu_{it}} = (\lambda_{it} - 1)dW_t$$

- wealth share positively exposed to $\sigma_w(\boldsymbol{y}_t)dZ_t^y$ iff $\gamma_{it} > \Gamma_t$
- drift in wealth shares if expectation error $dW_t$ drifts
- no contribution of $dZ_t^\gamma$: shocks to constraints/robustness parameters do not redistribute

## dynamics of risk premia

**Proposition 7:**

The wealth-weighted average multiplier evolves as

$$d\Gamma_t = \frac{\Delta_t}{\Gamma_t} dW_t + \boldsymbol{\nu}_t' d\boldsymbol{\gamma}_t$$

- two forces: model revisions $\boldsymbol{\nu}_t' d\boldsymbol{\gamma}_t$ and redistribution $\frac{\Delta_t}{\Gamma_t} dW_t$
- redistribution only works if there is heterogeneity, $\Delta_t > 0$
- redistribution suppresses risk premia after positive output shocks
- redistribution has a drift if expectation error $dW_t$ does

# Examples

# model revision shocks and redistribution

Consider a simple economy:

- one Lucas tree, output evolves as $\frac{dy_t}{y_t} = \mu dt + \sigma dZ_t^y$
- two agents: one with time-varying $d\gamma_t = \mu_\gamma(\gamma_t)dt + \sigma_\gamma(\gamma_t)dZ_t^\gamma$, one with a fixed $\gamma_t = \hat{\gamma}$
- two states: exogenous multiplier $\gamma_t$ and endogenous wealth share $\nu_t$ of the first agent
- weighted average $\Gamma(\gamma_t, \nu_t) = \hat{\gamma} + \nu_t(\gamma_t - \hat{\gamma})$
- possibly redistributive taxes to induce stationarity

# special case: model revision only

Special case: let $\nu_t = 1$ (agent with time-varying $\gamma_t$ takes over)

Interest rate:

$$r(\gamma_t) = \rho + \mu - \frac{\sigma^2}{\gamma_t}$$

Nests known exogenous interest rate specifications:

- Vasicek (1977) characterization
- Cox, Ingersoll, and Ross (1985) characterization

# special case: redistribution only

Suppose $\gamma_t$ are both constant: $\overline{\gamma}$ and $\underline{\gamma}$

- agent with higher $\gamma$ levers up to bet on output growth
- positive shocks make her relatively richer, suppress risk premium
- define risk premium as $\pi_t \equiv \dfrac{\sigma^2}{\Gamma_t}$, also given by $\pi_t = \rho + \mu - r_t$
- becomes a Markov process on $[\sigma^2/\overline{\gamma}, \sigma^2/\underline{\gamma}]$: characterization

$$\frac{d\pi_t}{\pi_t} = \mu_\pi(\pi_t)dt + \sigma_\pi(\pi_t)dZ_t^y$$

- $\mu_\pi(\cdot) < 0$: risk premium drifts down as the optimist makes money on average
- $\sigma_\pi(\cdot) < 0$: risk premium falls after positive growth surprises as the optimist bets on growth
- relates to Caballero and Simsek (2020) and Kekre and Lenel (2022)

# Objective and subjective premia

# expected excess returns

Take the market return $dR_t = \frac{1}{\sum_j s_j p_j(x_t)} \sum_j s_j p_j(x_t) dR_{jt}$

Expected returns under the physical measure $\mathbb{P}$ and $\mathbb{Q}_t$:

$$\mathbb{E}[dR_t] = \frac{|\sigma_w(\boldsymbol{y}_t)|^2}{\Gamma_t} dt$$
$$\mathbb{E}^{\mathbb{Q}_t}[dR_t] = |\sigma_w(\boldsymbol{y}_t)|^2 dt$$

Piazzesi, Salomao, and Schneider (2015), Nagel and Xu (2023):

- subjective premia less volatile, less responsive to business cycle

Premia constant with models (log utility), model selection moves objective premia in equilibrium

# estimation

Survey data: same idea as Bhandari, Borovička, and Ho (2025), Szőke (2022)

Observables: year-ahead excess return forecasts $f_t^{\text{sub}}$ and $f_t^{\text{obj}}$, realized returns $rx_{t+4}$

Three equations:

$$\log(f_t^{\text{sub}}) - \log(f_t^{\text{obj}}) = \log(\Gamma_t) + \alpha + \sigma_1 \epsilon_{1t} \qquad \text{(expectation gap)}$$

$$rx_{t+4} - f_t^{\text{sub}} = \underbrace{\frac{1-\Gamma_t}{\Gamma_t} \cdot f_t^{\text{sub}}}_{\text{drift}} + \sigma_2 \epsilon_{2t} \qquad \text{(forecast error)}$$

$$\Delta\Gamma_{t+1} = \underbrace{(1-\Gamma_t)\frac{rx_{t+4} - f_t^{\text{sub}}}{4}}_{\text{redistribution} \equiv \Delta\kappa_{1,t+1}} + \underbrace{\mu + \sigma_3 \epsilon_{3t}}_{\text{revisions} \equiv \Delta\kappa_{2,t+1}} \qquad \text{(evolution)}$$

Run MLE over $\{\Gamma_0, \alpha, \mu, \sigma_1, \sigma_2, \sigma_3\}$, recover $\{\Delta\Gamma_t, \Delta\kappa_{1t}, \Delta\kappa_{2t}\}$

# backing out $\Gamma_t$

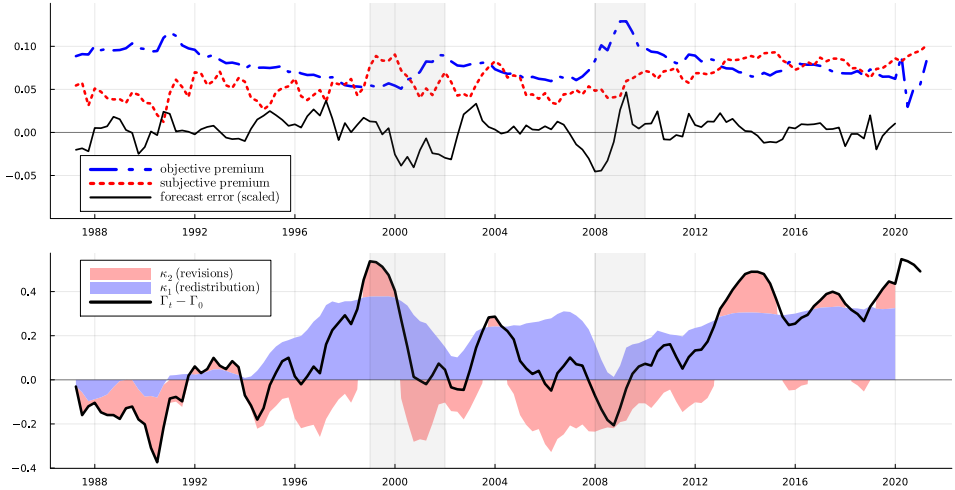



Figure: Recovered $\Gamma_t - \Gamma_0$ and the decomposition into redistribution and model revision parts.

# redistribution and intermediary capital

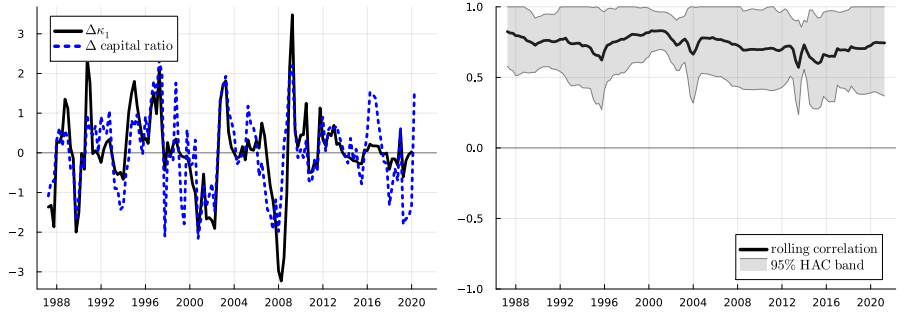



Figure: Recovered redistributive part of $\Delta\Gamma_{t+1}$ and the intermediary capital ratio from He, Kelly, and Manela (2017) on the left; rolling window correlation (40 quarters) on the right.

- correlation of $\Delta$ capital ratio with revisions $\sim 0.1$
- $\Delta$ capital ratio is a priced shock in data and model, revisions not priced in the model

# conclusion

An economy with model revisions $\longrightarrow$ dynamic risk limits

Simple aggregation in general equilibrium

- one scalar summary of wealth distribution determines asset prices and risk premia
- forecast errors from the "representative" model determine all dynamics
- clean separation of risk premium shocks into model revisions and redistribution

Identifiable from the gap between objective and subjective premia

- redistribution captures dynamics of intermediary equity

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# heuristic explanation

back

Take some $(L_t, \alpha_t)$:

$$\mathbb{P}\{\theta_t' dR_t \leq -\sqrt{L_t dt}\} \leq \alpha_t$$

Equivalently,

$$\Phi\left(-\frac{\sqrt{L_t dt} + \theta_t' \mu_R(x_t) dt}{\sqrt{\theta_t' \sigma_R(x) \sigma_R(x)' \theta_t dt}}\right) \leq \alpha_t$$

Suppose $\alpha \leq 1/2$, in the limit $dt \longrightarrow 0$,

$$\theta_t' \sigma_R(x_t) \sigma_R(x_t)' \theta_t \leq \frac{L_t}{(\Phi^{-1}(\alpha_t))^2}$$

With $L_t = \theta_t' \mu_R(x_t)$ and $\alpha_t = \Phi(-\sqrt{1/\gamma_t})$,

$$\mathbb{V}_t[\theta_t' dR_t] = \theta_t' \sigma_R(x_t) \sigma_R(x_t)' \theta_t dt \leq \gamma_t \cdot \theta_t' \mu_R(x_t) dt = \gamma_t \mathbb{E}_t[\theta_t' dR_t]$$

# recursive problem formulation

With $(w, x)$ as states, value $V(w, x)$ solves

back

$$\rho V(w,x) = \max_{c,\theta} \rho \log(c) + (r(x)w - c + w\theta'\mu_R(x))V_w(w,x) + \frac{\theta'\sigma_R(x)\sigma_R(x)'\theta}{2}V_{ww}(w,x)$$
$$+ \mu_x(x)'V_{x'}(w,x) + \frac{1}{2}\text{tr}[\sigma_x(x)'V_{xx'}(w,x)\sigma_x(x)] + w\theta'\sigma_R(x)\sigma_x(x)'V_{wx'}(w,x)$$

s.t. $\theta'\sigma_R(x)\sigma_R(x)'\theta \leq \gamma \cdot \theta'\mu_R(x)$

# extensions

Simple portfolios survive with income from outside of financial markets:

- taxes (inducing stationarity)
- perpetual youth of Yaari (1965), Blanchard (1985)

Key to preserve consumption and portfolio choice: additional terms linear in own wealth

$$dw_t = (r(x_t)w_t - c_t)dt + \theta_t' dR_t - \underbrace{w_t \varsigma(x_t) dt}_{\text{deterministic tax}} - \underbrace{w_t \tau(x_t)' dZ_t}_{\text{stochastic tax}}$$

Can handle any deterministic tax $\varsigma(x_t)$, stochastic "profit" taxes $\tau(x_t)' \propto \theta(x_t)' \sigma_R(x_t)$ result back

# relation to recursive preferences

Take Kreps and Porteus (1978) preferences in Duffie and Epstein (1992) form, keep EIS=1: back

$$V_t = \mathbb{E}_t \int_t^\infty \varphi(c_s, V_s) ds \quad \text{with} \quad \varphi(c, v) = \frac{\rho v(\gamma - 1)}{\gamma} \left[ \log(c) - \frac{\gamma}{\gamma - 1} \log\left( \frac{v(\gamma - 1)}{\gamma} \right) \right]$$

Value is no longer separable over $w$ and $x$:

$$V(w, x) = \frac{(w\eta(x))^{1-1/\gamma}}{1 - 1/\gamma}$$

Optimal portfolio includes hedging motives if $\gamma \neq 1$:

$$c^*(w, x) = \rho w$$

$$\theta^*(w, x) = \gamma \cdot [\sigma_R(x)\sigma_R(x)']^{-1} \mu_R(x) + (\gamma - 1)[\sigma_R(x)\sigma_R(x)']^{-1} \underbrace{\sigma_R(x)\sigma_x(x)' \frac{\eta_{x'}(x)}{\eta(x)}}_{\text{hedging motives}}$$

# multiplier problem

Let $\{Z_t\}_{t\geq 0}$ be a standard Brownian on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$, take an adapted process $\{h_t\}_{t\geq 0}$

- consider an adapted process $\{M_t\}_{t\geq 0} : M_0 = 1$ and $dM_t = -h_t M_t dZ_t$
- defines a probability measure $\mathbb{Q} : \mathbb{E}^{\mathbb{Q}}[\xi_t] = \mathbb{E}^{\mathbb{P}}[M_t \xi_t]$ for all bounded $\{\xi_t\}_{t\geq 0}$ and all $t \geq 0$
- $\{B_t\}_{t\geq 0}$ with $B_0 = 0$ and $dB_t = dZ_t - h_t dt$ is a standard Brownian on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{Q})$
- given an adapted process $\{\psi_t\}_{t\geq 0}$ and $m_t \equiv \log(M_t)$, agent solves a multiplier problem

$$\max_{\{c_t, \theta_t\}} \inf_{\mathbb{Q}} \mathbb{E}^{\mathbb{Q}} \left[ \int_0^\infty \rho e^{-\rho t} \log(c_t) dt + \int_0^\infty e^{-\rho t} \psi_t dm_t \right]$$

solving the problem back

# solving the multiplier problem

back

Log-likelihood process $m_t$ evolves as

$$dm_t = -\frac{1}{2}|h_t|^2 dt - h_t' dZ_t = \frac{1}{2}|h_t|^2 dt - h_t' dB_t$$

Recursive formulation:

$$\begin{aligned}\rho V(w,x) = \max_{c,\theta} \min_{h}\ & \rho \log(c) + \frac{\psi |h|^2}{2} \\ & + (r(x)w - c + w\theta'(\mu_R(x) - \sigma_R(x)h))V_w(w,x) + \frac{1}{2}\theta'\sigma_R(x)\sigma_R(x)'\theta V_{ww}(w,x) \\ & + \mu_x(x)' V_{x'}(w,x) + \frac{1}{2}\mathrm{tr}[\sigma_x(x)' V_{xx'}(w,x)\sigma_x(x)] + w\theta'\sigma_R(x)\sigma_x(x)' V_{wx'}(w,x)\end{aligned}$$

Separability preserved: $V(w,x) = \log(w) + \hat{\eta}(x)$ and

$$\begin{aligned} c^*(w,x) &= \rho w \\ \theta^*(w,x) &= \frac{\psi}{\psi+1} \cdot [\sigma_R(x)\sigma_R(x)']^{-1}\mu_R(x)\end{aligned}$$

# relation to standard robustness setup

back

In the standard case, model for states is misspecified too:

$$dx_t = \mu_x(x_t)dt + \sigma_x(x_t)dZ_t \equiv (\mu_x(x_t) - \sigma_x(x_t)h_t)dt + \sigma_x(x_t)dB_t$$

Recursive formulation:

$$\begin{aligned}\rho V(w,x) = &\max_{c,\theta} \min_{h} \rho \log(c) + \frac{\psi|h|^2}{2} \\ &+ (r(x)w - c + w\theta'(\mu_R(x) - \sigma_R(x)h))V_w(w,x) + \frac{1}{2}\theta'\sigma_R(x)\sigma_R(x)'\theta V_{ww}(w,x) \\ &+ (\mu_x(x) - \underbrace{\sigma_x(x)h}_{\text{new}})'V_{x'}(w,x) + \frac{1}{2}\text{tr}[\sigma_x(x)'V_{xx'}(w,x)\sigma_x(x)] + w\theta'\sigma_R(x)\sigma_x(x)'V_{wx'}(w,x)\end{aligned}$$

Separability $V(w,x) = \log(w) + \hat{\eta}(x)$ preserved but optimal $h$ and $\theta$ pick up $V_{x'}(w,x)$:

$$\theta^*(w,x) = \frac{\psi}{\psi+1} \cdot [\sigma_R(x)\sigma_R(x)']^{-1}\mu_R(x) - \frac{1}{\psi+1}[\sigma_R(x)\sigma_R(x)']^{-1}\sigma_R(x)\sigma_x(x)'\frac{\hat{\eta}_{x'}(x)}{\hat{\eta}(x)}$$

# stochastic taxes proportional to profits

back

Consider the following class of tax rates:

$$\tau(x_t) = \zeta(x_t)\gamma_t \cdot \sigma_R(x_t)'[\sigma_R(x_t)\sigma_R(x_t)']^{-1}\mu_R(x_t)$$

Tax payments proportional to resulting profits:

$$\tau(x_t)'dZ_t = \zeta(x_t)\theta(x_t)'\sigma_R(x_t)dZ_t = \zeta(x_t)\theta(x_t)'(dR_t - \mu_R(x_t)dt)$$

Optimal portfolio the same unless $\zeta(x_t)$ very negative:

$$\theta(w_t, x_t) = \min\{\gamma_t, 1 + \zeta(x_t)\gamma_t\} \cdot [\sigma_R(x_t)\sigma_R(x_t)']^{-1}\mu_R(x_t)$$

# returns

Vector of excess returns:

back

$$
\begin{aligned}
dR_t &\equiv \mu_R(x_t)dt + \sigma_{R,y}(x_t)dZ_t^y + \sigma_{R,\gamma}(x_t)dZ_t^\gamma \\
&= D(\boldsymbol{p}_t)^{-1}(\mu_p(x_t) + \boldsymbol{y}_t - r(x_t)\boldsymbol{p}(x_t))dt + D(\boldsymbol{p}_t)^{-1}(\sigma_{p,y}(x_t)dZ_t^y + \sigma_{p,\gamma}(x_t)dZ_t^\gamma)
\end{aligned}
$$

# PDE for asset prices

**Corollary:** the PDE for asset prices is linear.

back

$$r(x_t)p_j(x_t) = y_{jt} + \mathcal{D}p_j(x_t)\Big[\mu_x(x_t) \underbrace{-\frac{1}{\Gamma_t}\sigma_{x,y}(x_t)\sigma_w(\boldsymbol{y}_t)'}_{\text{risk adjustment}}\Big] + \frac{1}{2}\text{tr}[\mathcal{H}p_j(x_t)\sigma_x(x_t)\sigma_x(x_t)']$$

# ”exogenous interest rates”: Vasicek 1977

## Proposition 8

Let $\kappa_r > 0$ and $\sigma_r$ be parameters and suppose the process for $\gamma_t$ is

$$\frac{d\gamma_t}{\gamma_t} = \left(\kappa_r + \frac{\sigma_r^2 \gamma_t^2}{\sigma^4}\right) dt + \frac{\sigma_r \gamma_t}{\sigma^2} dZ_t^\gamma$$

with a reflecting boundary at $\gamma_t = 1$. The process for the interest rate is that of Vasicek (1977):

$$dr_t = \kappa_r(\rho + \mu - r_t)dt + \sigma_r dZ_t^\gamma$$

with a reflecting boundary at $r_t = \rho + \mu - \sigma^2$.

back

# ”exogenous interest rates”: Cox Ingersoll Ross 1985

**Proposition 9**

Let $\kappa_r > 0$ and $\sigma_r$ be parameters and suppose the process for $\gamma_t \in [\sigma^2/(\rho+\mu), 1]$ is

$$\frac{d\gamma_t}{\gamma_t} = \left(\kappa_r + \frac{\sigma_r^2\gamma_t^2}{\sigma^4}\left(\rho + \mu - \frac{\sigma^2}{\gamma_t}\right)\right)dt + \frac{\sigma_r\gamma_t}{\sigma^2}\sqrt{\rho+\mu-\frac{\sigma^2}{\gamma_t}}dZ_t^\gamma$$

with a reflecting boundary at $\gamma_t = 1$. The interest rate follows Cox, Ingersoll, and Ross (1985):

$$dr_t = \kappa_r(\rho + \mu - r_t)dt + \sigma_r\sqrt{r_t}dZ_t^\gamma$$

with a reflecting boundary at $r_t = \rho + \mu - \sigma^2$.

back

## redistribution shocks generate ”risk premium shocks”

**Proposition 9**

The drift and volatility of the risk premium are

$$\mu_\pi(\pi_t) = \left(\frac{\sigma}{\overline{\pi}\underline{\pi}}\right)^2 \cdot (\sigma^2(\overline{\pi} + \underline{\pi} - \pi_t) - \overline{\pi}\underline{\pi}) \cdot (\overline{\pi} - \pi_t)(\pi_t - \underline{\pi}) < 0$$

$$\sigma_\pi(\pi_t) = -\frac{\sigma}{\overline{\pi}\underline{\pi}} \cdot (\overline{\pi} - \pi_t)(\pi_t - \underline{\pi}) < 0$$

back